

Vishnu P R

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github.com/vishnupr01 | leetcode.com/u/vpr6748

Professional Summary

Software Engineer with 2+ years of experience building scalable backend systems using Node.js, Express.js, and RESTful APIs. Expert in AWS cloud infrastructure (EC2, S3, RDS), Docker containerization, Redis caching, and real-time communication using WebSockets and Socket.IO. Proficient in Java with Spring Boot, MVC architecture, microservices design, and database optimization. Proven track record of reducing API latency by 75% and implementing event-driven architectures for high-traffic platforms. Experienced in collaborating with mobile development teams (Swift, Kotlin) to design robust API contracts.

Technical Skills

- **Programming Languages:** JavaScript, TypeScript, Java, C, C++, SQL
- **Backend Frameworks:** Node.js, Express.js, Spring Boot, Spring MVC
- **Frontend Technologies:** React, Redux Toolkit, React Router, React Native, HTML5, CSS3, Tailwind CSS, Bootstrap
- **Databases:** MongoDB, MySQL, PostgreSQL, Redis, Firebase Firestore
- **Cloud and DevOps:** AWS (EC2, S3, RDS, CloudWatch, Lambda), Google Cloud Platform (GCP), Docker, Kubernetes, Nginx, PM2, Jenkins, CI/CD, Git, GitHub, Linux
- **APIs and Networking:** RESTful API, GraphQL, WebSockets, Socket.IO, HTTP/HTTPS, JWT, OAuth 2.0, API Gateway, Postman
- **Architecture Patterns:** MVC, Clean Architecture, Microservices, Event-Driven Architecture, SOLID Principles, Design Patterns
- **Real-Time Communication:** Socket.IO, WebRTC, Redis Pub/Sub, Server-Sent Events (SSE)
- **Additional Tools:** Mongoose, Sequelize, BullMQ, Message Queue, Axios, Cloudinary, Firebase, Razorpay, Nodemailer, Jest, Mocha

Professional Experience

Software Engineer

Meetmux

April 2025 – Present

Bangalore, India

Real-Time Communication Systems

- Designed real-time chat system using Socket.IO supporting 10,000+ concurrent users with event-driven architecture
- Built socket notification system for messaging and in-app notifications achieving sub-100ms latency
- Implemented group communication with automatic group creation and membership handling features
- Enabled cross-instance socket communication using Redis Pub/Sub for seamless horizontal scaling

Backend Performance and Scalability

- Optimized home feed API reducing response time by 75% from 400ms to 100ms through query optimization
- Implemented database indexing strategies improving query efficiency under high-traffic conditions
- Refactored backend modules improving code maintainability using SOLID principles and clean architecture
- Implemented BullMQ and Redis worker system for asynchronous job processing including post expiration
- Improved system reliability by 40% by decoupling heavy operations using message queue architecture

Backend Features and Platform Logic

- Designed backend menu and navigation logic with role-based access control (RBAC) and permission management
- Built user suggestion algorithm based on interests and geographic proximity using latitude/longitude distance calculation
- Implemented interest-based post interaction workflow enabling connection requests with notification triggers
- Developed secure RESTful APIs for authentication, content management, and social interactions

Authentication and Security

- Implemented multi-factor authentication combining face verification and OTP validation using Face Recognition API
- Built Firebase Cloud Messaging (FCM) notification system for push notifications validated via Postman
- Implemented JWT-based authentication with refresh token rotation and bcrypt password hashing

Cloud Infrastructure and Deployment

- Led backend deployment with AWS EC2 provisioning, auto-scaling, load balancing, and Nginx reverse proxy
- Configured AWS S3 for media storage, Amazon RDS for relational data, and Redis ElastiCache for caching
- Performed production debugging using AWS CloudWatch logs with monitoring dashboards ensuring 99.9% uptime
- Collaborated with Android (Kotlin) and iOS (Swift) teams to align API contracts and standardize responses

Key Achievements

- Reduced critical feed API latency by 75% (400ms to 100ms) improving user experience and engagement
- Enabled horizontally scalable real-time communication supporting 10,000+ concurrent connections
- Improved backend reliability by 40% through event-driven architecture and background job processing
- Took ownership of deployment and infrastructure for production platform with 50,000+ active users
- Application: [MeetMux on App Store](#)

Personal Projects

WE TALKS | Social Media Platform

- Developed full-stack social platform using MERN stack with scalable codebase contributing to 25% user retention
- Engineered real-time video communication using WebRTC and Socket.IO resulting in 40% engagement increase
- Deployed backend on AWS EC2 with PM2 achieving 30% reduction in API response time
- Managed media storage with Cloudinary CDN increasing data retrieval speeds by 25%
- Developed responsive frontend using React and Tailwind CSS with Redux Toolkit for state management
- Technologies: MongoDB, Express.js, React, Node.js, Tailwind CSS, Redux Toolkit, Socket.IO, WebRTC, AWS, Cloudinary
- GitHub: [Frontend](#) | [Backend](#)

VKart | E-Commerce Platform for Electronics

- Created full-stack eCommerce website for electronics with 3 product categories completed in 1 month
- Implemented secure authentication with Nodemailer OTP verification enhancing account security
- Engineered shopping cart and checkout workflow supporting 3 payment methods through Razorpay

gateway

- Built server-side rendered frontend with EJS templates and Bootstrap ensuring responsive design
- Technologies: Node.js, Express.js, MongoDB, EJS, Bootstrap, Razorpay, Nodemailer, Session Management
- GitHub: [Repository](#)

Task Management System | Clean Architecture Application

- Designed task management app using MERN stack and TypeScript with role-based authentication
- Implemented CRUD operations and task reassignment with React Calendar for visual scheduling
- Built responsive user interface with React ensuring optimal usability across all devices
- Developed RESTful backend API with Node.js and MongoDB ensuring efficient data handling
- Technologies: MongoDB, Express.js, React, Node.js, TypeScript, React Calendar, JWT Authentication
- GitHub: [Frontend](#) | [Backend](#)

Image Stocker | Image Management Application

- Developed image management app using MERN stack and TypeScript with JWT authentication
- Implemented drag-and-drop interface using React DnD for seamless image reordering
- Integrated Cloudinary cloud storage for scalable image management with fast uploads
- Provided image management features including title editing, reordering, and deletion with responsive frontend
- Technologies: MongoDB, Express.js, React, Node.js, TypeScript, React DnD, Cloudinary API, JWT
- GitHub: [Repository](#)

Education

Certificate in MERN Full Stack Development
Prototype

2023 – 2024
Kozhikode, Kerala

- Certificate: [View Certificate](#)

Diploma in Computer Science and Engineering
IPT and GPTC Shoranur

2018 – 2021
Kerala